INTERIM MILITARY SPECIFICATION

CLEANING COMPOUND, COFFEE URN (WATER COMPARTMENT)

This interim Federal Specification was developed by U. S. Army Natick Laboratories (GL), Natick, Mass., based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers one type and grade of a coffee urn cleaning compound (see 6.1).

2. APPLICABLE SPECIFICATIONS, STANDARDS AND OTHER PUBLICATIONS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FEDERAL SPECIFICATIONS

L-P-378 —Plastic Film (Polyethylene Thin Gage)
PPP-D-723—Drums, Fiber

FEDERAL STANDARDS

Fed. Std. No. 102—Preservation, Packaging and Packing Levels
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies)

(Activities outside the Federal Government may obtain copies of Federal Specifications and standards as outlined under General Information in the Index of Federal Specifications and Standards at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Auburn, Washington.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

MILITARY STANDARDS

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129—Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invita-

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tion for bids or request for proposal shall apply.

TOILET GOODS ASSOCIATION INC.
Specification No. 28—Magnesium Oxide

(Application for copies should be addressed to the Toilet Goods Association Inc., 1270 Avenue of the Americas, New York 20, N.Y.)

AMERICAN CHEMICAL SOCIETY
Specification for Reagent Chemicals

(Application for copies should be addressed to the American Chemical Society, 1155 16th Street, N.W., Washington 6, D. C.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 **Material.** The cleaning compound shall be a dry blend consisting only of sulfamic acid, magnesium oxide, 1,3 diethylthiourea and when required methyl violet. No other ingredients shall be incorporated into the compound.

3.1.1 *Sulfamic acid.* The sulfamic acid shall be a technical grade in crystalline form and shall assay not less than 99.0 percent.

3.1.2 *Magnesium oxide.* The magnesium oxide shall conform to Toilet Goods Association Specification No. 28.

3.1.3 *1,3 Diethylthiourea ($C_2H_5NHCSNHC_2H_5$).* The 1,3 diethylthiourea shall be a commercial product having a melting point not less than 70° C.

3.1.4 *Methyl violet (when required).* Methyl violet, when required, shall be a technical grade.

3.2 **Composition.**

3.2.1 *Sulfamic acid.* The cleaning compound shall contain not less than 96.0 percent by weight of sulfamic acid when tested as specified in 4.4.2.

3.2.2 *Magnesium oxide.* The cleaning compound shall contain 1.0 ± 0.2 percent by weight of magnesium oxide when tested as specified in 4.4.3.

3.2.3 *1,3 Diethylthiourea.* The cleaning compound shall contain not less than 1.0 percent by weight of 1,3 diethylthiourea.

3.3 **Color.** The color of the cleaning compound shall be other than white so that it will not be mistaken for salt or sugar. If a colorant is required, the product shall be tinted by the use of not more than 0.05 percent by weight powdered methyl violet.

3.4 **Workmanship.** The material shall be thoroughly mixed. The finished product shall be clean, free-flowing and shall not lump or cake. The quality shall conform to the levels established in this specification.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Inspection.* Inspection shall be in accordance with Standard MIL–STD–105, except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of not more than 1,000 containers of the cleaning compound.

4.2 **Inspection of materials and components.** The supplier is responsible for insuring that all materials and components used in the end item were tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern. Conformance to the requirements listed in table I will be acceptable upon supplier's certification of compliance thereto.

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TABLE I. *Requirements acceptable by supplier's certificate of compliance.*

Material	Requirement
Sulfamic acid	Technical grade; crystalline form; assay not less than 99.0 percent.
1,3 Diethylthiourea	Melting point not less than 70°C.; cleaning compound composition contains not less than 1.0 percent by weight.
Methyl violet (when used)	Technical grade; not more than 0.05 percent by weight.

4.3 Inspection of the end item.

4.3.1 *Examination of the end item.* The end item shall be examined for the defects in 4.3.2 at inspection level S–4 and acceptable quality level (AQL) 4.0 percent defective. Random samples shall be drawn from each lot of end items offered for inspection. The lot size for purpose of determining the sample size in accordance with Standard MIL–STD–105, shall be expressed in units of filled containers of the specified capacity.

4.3.2 *Examination of the container for defects in construction, markings, color, net contents and workmanship.* The sample unit for this examination shall be one filled container.

Examine	*Defect*
Construction	Not in container specified.
	Any leakage of contents.
	Any split, break, dent, hole, tear, puncture, or other defect which may affect serviceability.
	Seams or closures not as specified in liner or container.
	Polyethylene liner missing.
Markings (label or printing).	Omitted, illegible, incorrect, incomplete, not in accordance with requirements.
	Label or printing not clear, smeared.
Color	White.
Net contents	Less than the specified or indicated quantity.
Workmanship	Not clean; any foreign matter.
	Not free flowing.
	Lumping or caking.

4.3.3 *Testing of the cleaning compound.* The cleaning compound shall be tested for sulfamic acid and magnesium oxide content as specified in 3.2 for each lot presented for examination. The sample unit for testing shall be a 2-pound composite of the cleaning compound from randomly selected containers and at random levels in the containers selected for examination of the end item (see 4.3.1). A minimum one-pound specimen shall be selected from each of the containers. The material selected from the individual containers shall be thoroughly mixed and the tests performed on the composite sample. Duplicate determinations shall be performed on the composite sample. The average value of both determinations shall meet the requirement for each test.

4.4 Test procedures.

4.4.1 *Composition.* The sulfamic acid and magnesium oxide content shall be determined on the composite sample (see 4.3.3). Analytical reagent grade chemicals and distilled water shall be used throughout the tests. Where applicable, the reagents shall meet the requirements of the American Chemical Society Specifications (Current Revision) for reagent chemicals.

4.4.2 *Sulfamic acid.*

4.4.2.1 *Reagents.*

(a) *Standard sodium nitrite solution, 0.25 normal.* Standardize against purified sulfamic acid. The sulfamic acid shall be purified according to the following procedure: Dissolve 125 grams of sulfamic acid in 300 grams of water previously heated to 70° C. Filter three times, with consequent lowering of temperature, discard the materials which crystallize from the solution. Cool the filtrate rapidly in an ice-salt mixture, and let stand 20 minutes. Remove the crystals thus formed by suction filtration. Wash the crystals with a small portion of ice water, then twice with cold ethanol and finally with ether. Air dry for 1 hour, grind

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in an agate mortar, and store in a desiccator.

(b) *Sulfuric acid.* Fifty percent by volume.

(c) *Potassium bromide-saturated solution.*

(d) *Crushed ice.*

4.4.2.2 *Apparatus.*

(a) *pH meter,* Beckman zeromatic or equivalent with a polarizing current facility.

(b) *Silver billet electrode,* Beckman No. 39261 or equal.

(c) *Platinum electrode* having one inch of exposed platinum.

(d) *Buret.* Fifty-milliliter capacity graduated on 0.1 milliliter divisions.

(e) *Volumetric flask.* 500-milliliter capacity.

(f) *Stirrer.*

(g) *Pipet.* 100-milliliter delivery.

4.4.2.3 *Procedure.* Dissolve a five-gram sample of the cleaning compound, weighed to the nearest milligram, in water, transfer to a 500-milliliter volumetric flask, and dilute to the mark with water. Use cool water throughout; the temperature shall not exceed 25° C. Transfer a 100-milliliter aliquot of the solution to a 400-milliliter beaker. Add 25 milliliters of sulfuric acid, 15 milliliters of potassium bromide solution, and 40 to 50 grams of crushed ice. Connect the electrodes to the pH meter using the polarizing jumper. The silver billet electrode shall be connected to the input jack and the platinum electrode to the reference jack. Set the meter to read MV. Agitate the solution with the stirrer. Immerse the active surfaces of the electrodes in the solution. Place the buret with the sodium nitrite solution so the tip is below the solution of the compound. Proceed with the titration by adding definite increments of sodium nitrite solution. The equivalence point shall be taken at the first permanent surge of the galvanometer. Calculate the percentage of sulfamic acid as follows:

$$\text{Sulfamic acid, percent} \pm 0.5 = \frac{\text{ml sodium nitrite} \times \text{normality} \times 48.5}{\text{weight of sample}}$$

4.4.3 *Magnesium oxide.*

4.4.3.1 *Reagents.*

(a) *Concentrated sulfuric acid.*

(b) *Concentrated nitric acid.*

(c) *Ammonium chloride.*

(d) *Ten-percent diammonium phosphate solution.*

(e) *Concentrated ammonium hydroxide.*

(f) *Five-percent ammonium hydroxide solution.*

(g) *Twenty-percent hydrochloric acid solution.*

(h) *Methyl purple or methyl orange pH indicator solutions.*

4.4.3.2 *Procedure.* Weigh 10 grams of the cleaning compound to the nearest milligram and transfer to a 400-milliliter beaker. Add 10 milliliters of concentrated sulfuric acid and heat to dense white fumes. Add 1 milliliter of concentrated nitric acid dropwise, allowing it to flow down the sides of the beaker. Heat again to fuming. Repeat the addition of nitric acid and the fuming until the solution becomes pale yellow or colorless. Allow to cool. Dilute to 100 milliliters with distilled water. Add 2 ± 0.5 grams of ammonium chloride and dissolve. Add 20 milliliters of 10 percent diammonium phosphate solution. Stir vigorously and neutralize with ammonium hydroxide to the methyl purple of methyl orange end point. Cool, add about a 10 percent excess of the ammonium hydroxide solution, and stir vigorously to initiate precipitation. Allow the precipitate to settle at least 4 hours. Filter through Whatman No. 42 filter paper and wash with 5 percent ammonium hydroxide. Char and ignite in a tared crucible at 1000° C. until constant weight is obtained. Calculate the percentage of magnesium oxide as follows:

$$\text{Magnesium oxide, percent} = \frac{\text{weight of residue} \times 36.2}{\text{weight of sample}}$$

5. PREPARATION FOR DELIVERY

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Standard Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packing for civilian agency procurements.

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 *Level A.* One hundred pounds of cleaning compound shall be packed in a fiber drum conforming to type II, grade A of Specification PPP-D-723. The interior of each drum shall be lined with a minimum 0.003-inch thick polyethylene bag fabricated from material conforming to type I, grade A, finish 1 of Specification L-P-378. Each bag shall be closed by heat sealing, allowing enough material for two additional heat seals. All excess air shall be removed prior to heat sealing.

5.1.2 *Level B.* One hundred pounds of cleaning compound shall be packed in a fiber drum conforming to type I, grade A of Specification PPP-D-723. Each drum shall be provided with a liner as specified in 5.1.1.

5.1.3 *Level C.* The cleaning compound shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.2 Marking.

5.2.1 *Civilian agencies.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Standard Fed. Std. No. 123.

5.2.2 *Military requirements.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Standard MIL-STD-129.

5.2.3 *Precautionary marking.* In addition to marking specified herein, each shipping container shall be marked with the following precautionary marking:

"CONCENTRATED ACID — KEEP COVER CLOSED, USE CAUTION WHEN HANDLING DRY OR IN SOLUTION.

WARNING:

CAUTION SHOULD BE OBSERVED IN HANDLING. WEAR GOGGLES TO PROTECT EYES; GLOVES ARE ALSO ADVISABLE. AVOID BREATHING DUST. IN CASE OF CONTACT WITH SKIN OR CLOTHING, FLUSH WITH WATER. IN CASE OF CONTACT WITH EYES, FLUSH WITH WATER AND GET MEDICAL ATTENTION."

6. NOTES

6.1 Intended use. The cleaning compound covered by this specification is intended for the removal of hard water scale from the water compartment of military type coffee urns. The recommended frequency is as follows: 60 to 90 days in areas with water hardness of approximately 90 parts per million (ppm) $CaCO_3$; 30 to 60 days in areas with water hardness of approximately 180 ppm $CaCO_3$. In addition the cleaning compound can be used for the removal of hard water deposits in steam tables and boilers.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable level of packing (see 5.1).

(c) Packing and marking requirements other than 5.2.

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